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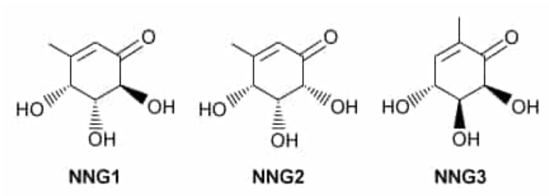


The preparation of these compounds involves hydroxyhalogenation and acetonide protection of the diol in Compound **1** to form an iodohydrin (Compound **2**). Compound **2** then undergoes an intramolecular S_N2 reaction to form an epoxide (not shown), which subsequently undergoes another S_N2 reaction with hydroxide. This is followed by the addition of the benzyl (Bz) protecting group and removal of the acetonide protecting group to form Compound **3**. Compound **3** is then oxidized using the oxidizing agent IBX to give a mixture of the desired product (Compound **4**) and an overoxidized product (Compound **5**).

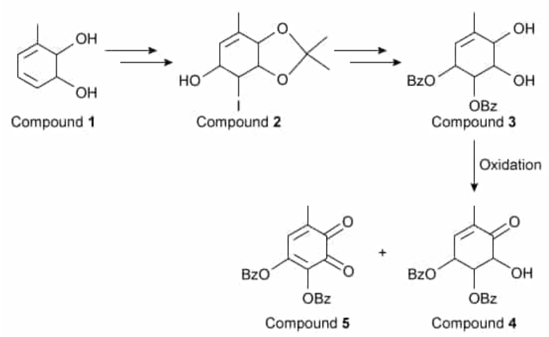
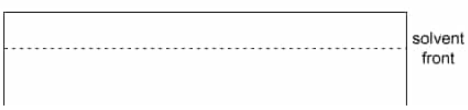


Figure 1 Selected steps in the synthesis of gabosine derivatives. (Note: Bz is an alcohol protecting group, and stereochemistry is not shown.)

Researchers studied different reaction conditions for the oxidation of Compound **3** by monitoring the reaction with thin-layer chromatography (TLC) on a silica plate with a nonpolar solvent. The results in Figure 2 show spots corresponding to Compound **4** and/or Compound **5**, depending on the reaction conditions. If the reaction did not go to completion, a spot corresponding to the starting material Compound **3** was also observed. After optimal oxidation conditions were determined, Compound **4** was converted to NNG3 by removal of the Bz protecting group. After its synthesis, a 1 g/mL sample of NNG3 in a 1 dm sample cell was found to have a specific rotation of −350° at 20 °C using 589 nm light.



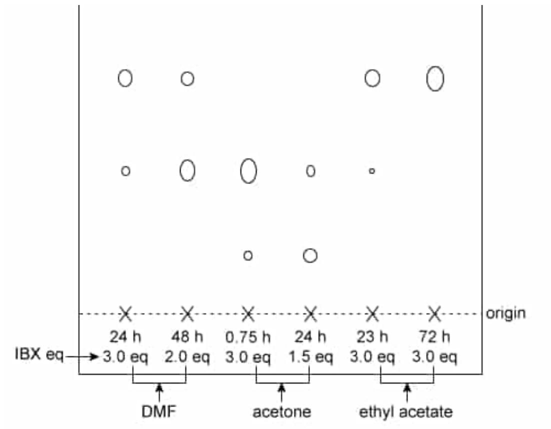
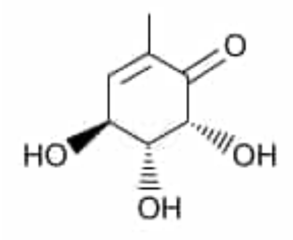


Figure 2 TLC results for oxidation reaction optimization with the reaction time, equivalents of the oxidizing agent IBX, and reaction solvent marked in each lane

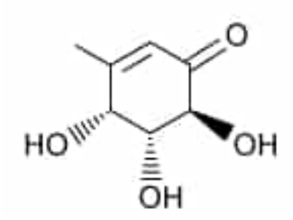
Which of the following structures is an epimer of NNG3?

- A.



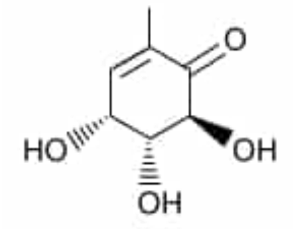
[15%]

- B.



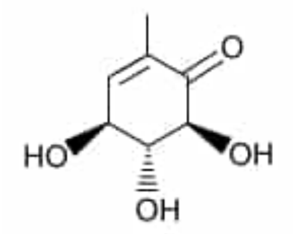
[8%]

- ✔ C.



[69%]

- D.

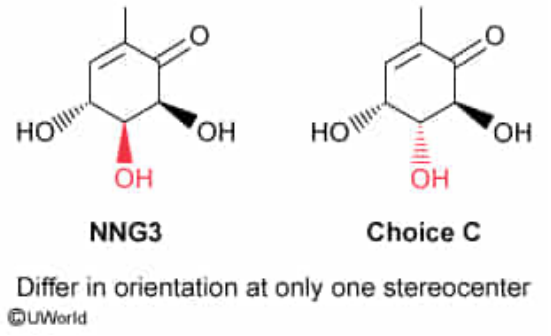


[6%]

Correct
69% answered correctly

Explanation:

Epimers



Stereoisomers are compounds with the same molecular formula and atom connectivity but with bonds oriented differently in space. Stereoisomers have one or more stereocenters, also known as chiral centers, that consist of an atom bonded to four different substituents. Diastereomers are a type of stereoisomer where at least one (but not all) of the stereocenters differ in orientation. Epimers are a type of diastereomer that **differ in spatial orientation** at only **one stereocenter**.

NNG3 has three stereocenters. Therefore, an epimer of NNG3 must have the same atom connectivity and number of stereocenters but differ in orientation at only one of the stereocenters. Choice C shows the only structure that fits these characteristics.

(Choice A) This structure differs in orientation from NNG3 at *all* the stereocenters and is the enantiomer of NNG3.

(Choice B) This is the structure of NNG1; it contains the same molecular formula as NNG3 but different atom connectivity (ie, the CH_3 group on the ring is bonded to a different carbon atom). Therefore, this structure is a constitutional isomer (or structural isomer) of NNG3 rather than an epimer.

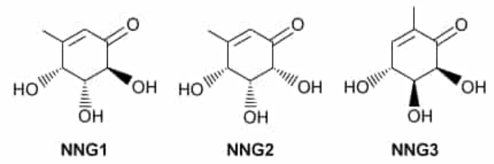
(Choice D) This structure differs in orientation from NNG3 at *two* stereocenters rather than one stereocenter. It is a *diastereomer* of NNG3 but not an epimer.

Educational objective:
Stereoisomers are compounds with the same molecular formula and atom connectivity but with bonds oriented differently in space. Epimers are a type of diastereomer that differ in spatial orientation at only one stereocenter.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403072
System:
Introduction to Organic Chemistry

Carbasugars, also known as pseudosugars, are carbohydrate analogues in which the ring oxygen has been replaced by a carbon atom. Gabosines and their non-natural derivatives (NNG1, NNG2, and NNG3) are carbasugars with potential biological activity, including antibiotic and anticancer properties, and therefore are desirable synthetic targets.



The preparation of these compounds involves hydroxyhalogenation and acetonide protection of the diol in Compound **1** to form an iodohydrin (Compound **2**). Compound **2** then undergoes an intramolecular S_N2 reaction to form an epoxide (not shown), which subsequently undergoes another S_N2 reaction with hydroxide. This is followed by the addition of the benzyl (Bz) protecting group and removal of the acetonide protecting group to form Compound **3**. Compound **3** is then oxidized using the oxidizing agent IBX to give a mixture of the desired product (Compound **4**) and an overoxidized product (Compound **5**).

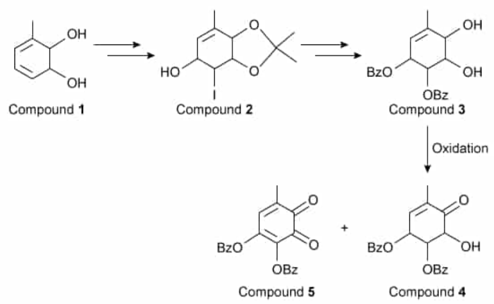


Figure 1 Selected steps in the synthesis of gabosine derivatives. (Note: Bz is an alcohol protecting group, and stereochemistry is not shown.)

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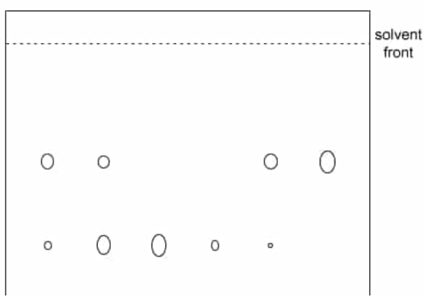


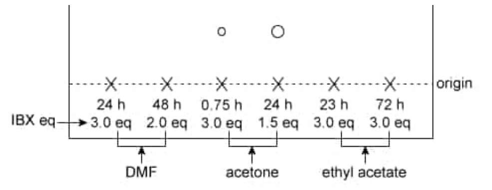
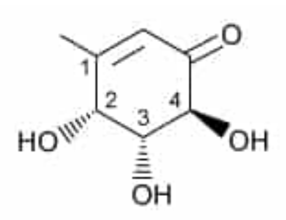


Figure 2 TLC results for oxidation reaction optimization with the reaction time, equivalents of the oxidizing agent IBX, and reaction solvent marked in each lane

NNG1 is shown below with certain carbon atoms labeled.

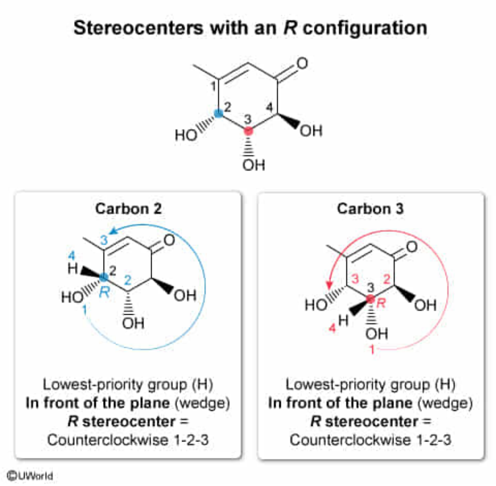


Which of the following carbon atoms are stereocenters with an *R* configuration?

- A. Carbon 1 only [3%]
- B. Carbon 4 only [14%]
- C. Carbons 2 and 3 only [64%]
- D. Carbons 2, 3, and 4 only [16%]

Correct
65% answered correctly

Explanation:



A chiral carbon is a type of stereocenter with four different substituents attached to a central carbon. The stereochemistry of the chiral carbon is designated as ***R* or *S* configuration** and depends on the spatial orientation of the substituents. The **priority** and **circular arrangement** of the substituents must both be considered to determine *R*/*S* configuration:

- The highest priority (ie, priority is 1) is assigned to the substituent atom with the greatest atomic number. If two substituent atoms are the same, then the atomic numbers of the next attached atoms are considered.

- When the **lowest-priority group** (ie, priority is 4) is pointed **behind the plane (dashed)**, a **clockwise** arrangement of groups 1, 2, and 3 gives an ***R* configuration** whereas a **counterclockwise** arrangement gives an ***S* configuration**.

- When the **lowest-priority group** (ie, priority is 4) is pointed **in front of the plane (wedged)**, a **clockwise** arrangement of groups 1, 2, and 3 gives an ***S* configuration** whereas a **counterclockwise** arrangement gives an ***R* configuration**.

Carbons 2 and 3 are chiral centers because each has four different substituents. The substituents with priority 1, 2, and 3 on carbons 2 and 3 are arranged in a counterclockwise configuration, and the lowest-priority group (H) is pointed in front of the plane (wedged) on carbons 2 and 3. Therefore, carbons 2 and 3 have only an *R* configuration.

(Choice A) The sp^2 hybridized carbons in an alkene can be stereocenters if the alkene can be classified as either cis/trans or *E/Z*. Carbon 1 is a stereocenter as it is part of a *Z* alkene, but its configuration cannot be assigned as *R* or *S*.

(Choices B and D) Carbon 4 is a chiral carbon because it has four different substituents. The substituents with priority 1, 2, and 3 are in a counterclockwise arrangement with the lowest-priority group pointed *behind* the plane; therefore, carbon 4 has an *S* configuration.

Educational objective:
To determine the *R*/*S* configuration of a chiral carbon, the priority of the substituents and their circular arrangement must be examined. When the lowest-priority group is pointed behind the plane (dashed), a clockwise arrangement of priorities 1, 2, and 3 is *R* whereas a counterclockwise arrangement is *S*.

When the lowest-priority group is pointed in front of the plane (wedged), a clockwise arrangement is *S* whereas a counterclockwise arrangement is *R*.

Time spent:0 sec	**QID**:403073
Subject: Organic Chemistry	**System:** Introduction to Organic Chemistry

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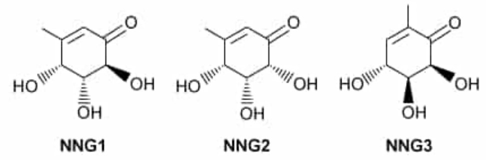

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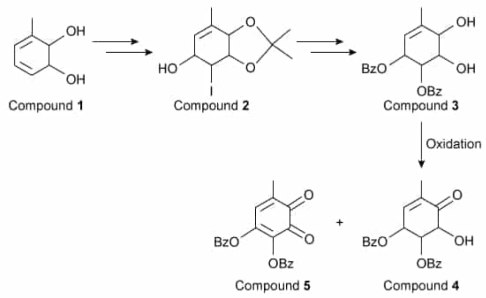

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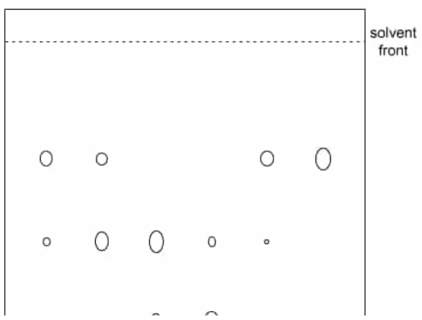


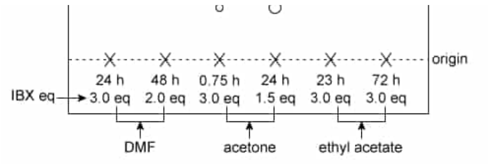
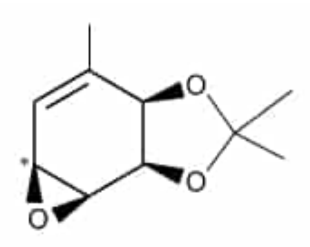


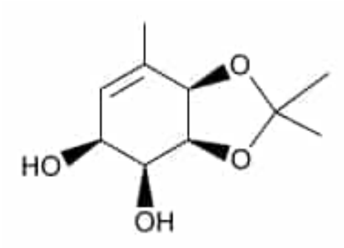
Figure 2 TLC results for oxidation reaction optimization with the reaction time, equivalents of the oxidizing agent IBX, and reaction solvent marked in each lane

The structure of the epoxide formed from Compound **2** is shown below.



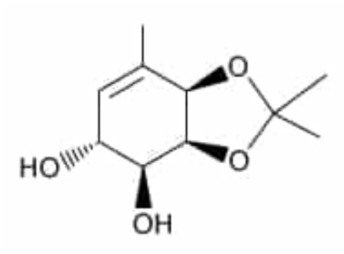
Based on the reaction described in the passage, what is the structure of the product when hydroxide reacts with the carbon marked by an asterisk, followed by protonation using HCl?

A.



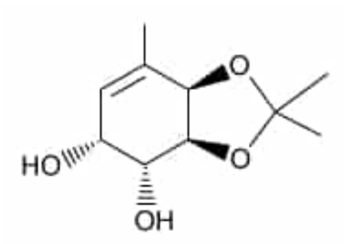
[10%]

B.



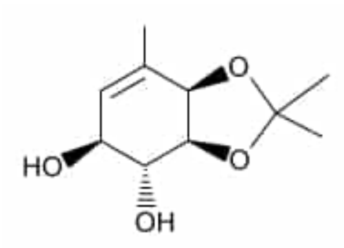
[67%]

C.



[14%]

D.



[7%]

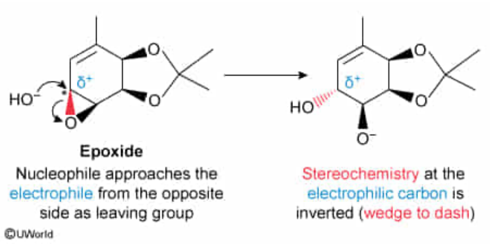
Correct

68% answered correctly

Explanation:

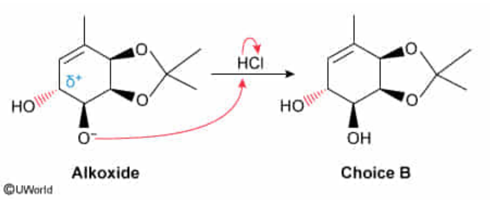
An S_N2 reaction is a **concerted substitution reaction**. A nucleophile attacks an electrophile on the opposite side of the leaving group (ie, the nucleophile does a **backside attack**), and if the electrophilic center is a chiral center, the **stereochemistry** of the chiral center is inverted.

The passage states that the epoxide shown in this question undergoes an S_N2 reaction. In this reaction, hydroxide (^-OH) is the nucleophile that attacks the carbon of the epoxide marked with the asterisk (ie, the electrophile). At the same time, the bond between the electrophile and the leaving group (ie, the oxygen of the epoxide) breaks, causing the epoxide to open and form an alkoxide.



Because the leaving group is pointed in front of the plane (wedged), the nucleophile approaches the electrophile from behind the plane (ie, opposite side of leaving group) and forms a bond with the electrophile that points behind the plane (dashed). Therefore, the electrophilic carbon undergoes inversion of stereochemistry (*S* to *R*, in this case). The carbon atom bonded to the alkoxide retains the stereochemistry of the epoxide (wedged) because it was not attacked by the nucleophile.

After nucleophilic attack, the alkoxide is protonated by an acid.



The structure in Choice B correctly shows the inversion of stereochemistry at the electrophilic carbon and the retention of stereochemistry of the leaving group.

(Choice A) This structure does not show the inversion of stereochemistry at the electrophilic carbon.

(Choice C) This structure incorrectly shows the inversion of stereochemistry at *both* epoxide carbons. Only the carbon attacked by the nucleophile undergoes inversion.

(Choice D) This structure shows the outcome of an S_N2 reaction if the nucleophile attacked the other carbon (not marked by an asterisk) in the epoxide.

Educational objective:
S_N2 reactions are concerted substitution reactions in which a nucleophile attacks an electrophile and the bond between the electrophile and the leaving group simultaneously breaks. The nucleophile does a backside attack, and if the electrophilic carbon is chiral, the stereochemistry of the chiral carbon is inverted.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403074
System:
Introduction to Organic Chemistry

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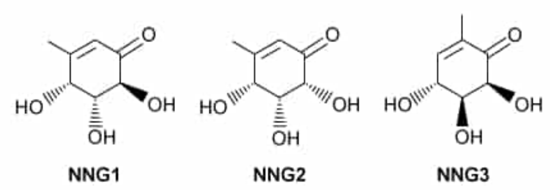

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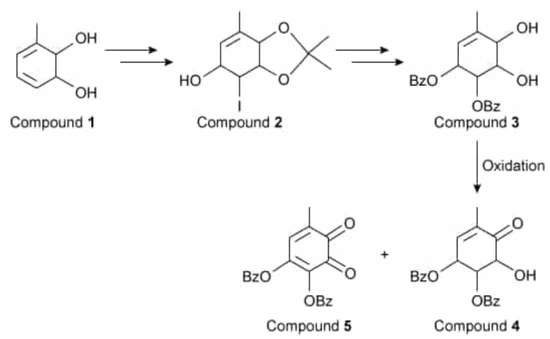

Figure 1 Selected steps in the synthesis of gabosine derivatives. (Note: Bz is an alcohol protecting group, and stereochemistry is not shown.)

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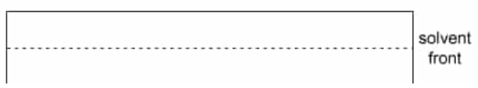


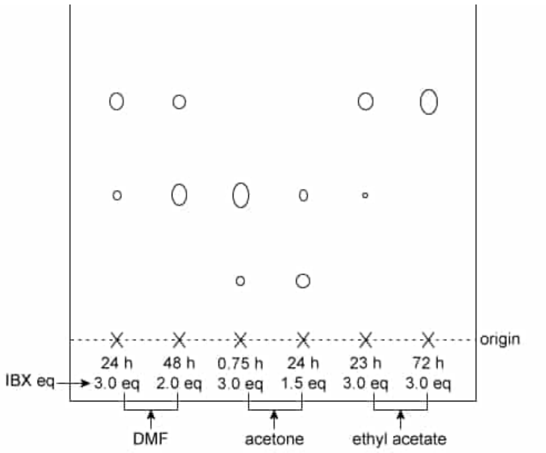


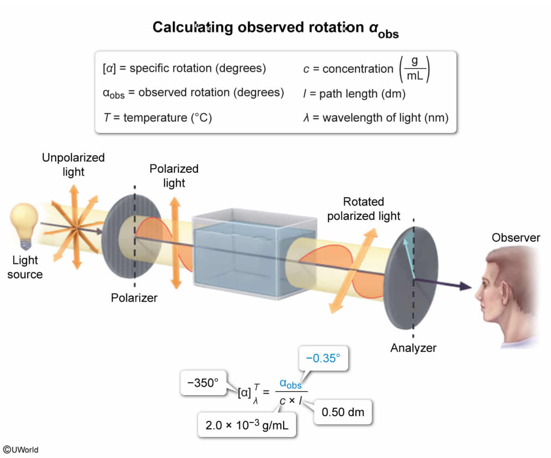
Figure 2 TLC results for oxidation reaction optimization with the reaction time, equivalents of the oxidizing agent IBX, and reaction solvent marked in each lane

Based on the specific rotation stated in the passage, what would be the observed rotation of NNG3 if a 2.0×10^{-3} g/mL sample in a 0.50 dm cell were used to take the measurement?

- A. −7.0° [6%]
- B. −1.4° [17%]
- C. −0.70° [25%]
- D. −0.35° [50%]

Correct
50% answered correctly

Explanation:



Specific rotation [α] describes the direction (+ or −) and magnitude (degrees) a chiral molecule rotates plane-polarized light. A clockwise rotation is classified as positive (+), and a counterclockwise rotation is classified as negative (−).

The **observed rotation** is **dependent upon sample concentration and path length** of the cell used (ie, a more concentrated sample or longer path length increases the observed rotation). Therefore, **[α] is standardized** to show what the rotation would be if the path length were 1 dm and the sample concentration were 1 g/mL. The value of [α] is calculated using the equation:

$$[\alpha]_{\lambda}^{T} = \frac{\alpha_{obs}}{c \times l}$$

where α_{obs} is the observed rotation (degrees), c is the sample concentration (g/mL), l is the path length of the cell used (dm), T is the temperature (°C), and λ is the wavelength of the light used (nm). Although T and λ affect a molecule's rotation, they are only reported with [α] and do not play a part in the specific rotation calculations. The units of specific rotation are (degrees · mL) / (g · dm) but results are usually expressed in just degrees for simplicity.

The observed rotation of NNG3 can be calculated from the [α] given in the passage and the concentration and path length given in the question by rearranging the above equation:

$$\alpha_{obs} = [\alpha] \times \mathrm{c} \times 1$$

$$\alpha_{obs} = \left(\frac{-350° \cdot \mathrm{mL}}{\mathrm{g} \cdot \mathrm{dm}}\right)\left(2.0 \times 10^{-3}\frac{\mathrm{g}}{\mathrm{mL}}\right)(0.50\ \mathrm{dm})$$

$$\alpha_{obs} = \left(\frac{-350° \cdot \cancel{\mathrm{mL}}}{\cancel{\mathrm{g} \cdot \mathrm{dm}}}\right)\left(1.0 \times 10^{-3}\frac{\cancel{\mathrm{g} \cdot \mathrm{dm}}}{\cancel{\mathrm{mL}}}\right)$$

$$\alpha_{obs} = -0.35°$$

(Choice A) A value of −7.0° is obtained if [α], c, and l are incorrectly multiplied by temperature (ie, the equation $\alpha_{obs} = [\alpha] \times c \times l \times T$ is used). Temperature does not play a part in the specific rotation calculations.

(Choice B) A value of −1.4° is obtained if [α] and c are divided by l instead of multiplied (ie, the equation $\alpha_{obs} = \frac{[\alpha] \times c}{l}$ is used).

(Choice C) A value of −0.70° is obtained if the path length of the sample cell is not taken into account (ie, the equation $\alpha_{obs} = [\alpha] \times c$ is used).

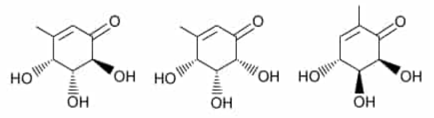
Educational objective:
Specific rotation [α] describes the direction (+ or −) and magnitude (degrees) by which a chiral molecule rotates plane-polarized light. The observed rotation α_{obs} is dependent upon sample concentration (c in g/mL) and path length (l in dm) of the cell used; temperature T (°C) and wavelength λ (nm) affect a molecule's rotation and are reported with [α]. Specific rotation [α] is standardized to account for c and l:

$$[\alpha]_{\lambda}^{T} = \frac{\alpha_{obs}}{c \times l}$$

Time spent:0 sec
Subject:
Organic Chemistry

QID:403075
System:
Introduction to Organic Chemistry

Carbasugars, also known as pseudosugars, are carbohydrate analogues in which the ring oxygen has been replaced by a carbon atom. Gabosines and their non-natural derivatives (NNG1, NNG2, and NNG3) are carbasugars with potential biological activity, including antibiotic and anticancer properties, and therefore are desirable synthetic targets.



The preparation of these compounds involves hydroxyhalogenation and acetonide protection of the diol in Compound **1** to form an iodohydrin (Compound **2**). Compound **2** then undergoes an intramolecular S_N2 reaction to form an epoxide (not shown), which subsequently undergoes another S_N2 reaction with hydroxide. This is followed by the addition of the benzyl (Bz) protecting group and removal of the acetonide protecting group to form Compound **3**. Compound **3** is then oxidized using the oxidizing agent IBX to give a mixture of the desired product (Compound **4**) and an overoxidized product (Compound **5**).

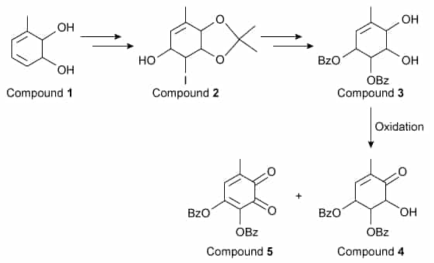


Figure 1 Selected steps in the synthesis of gabosine derivatives. (Note: Bz is an alcohol protecting group, and stereochemistry is not shown.)

Researchers studied different reaction conditions for the oxidation of Compound **3** by monitoring the reaction with thin-layer chromatography (TLC) on a silica plate with a nonpolar solvent. The results in Figure 2 show spots corresponding to Compound **4** and/or Compound **5**, depending on the reaction conditions. If the reaction did not go to completion, a spot corresponding to the starting material Compound **3** was also observed. After optimal oxidation conditions were determined, Compound **4** was converted to NNG3 by removal of the Bz protecting group. After its synthesis, a 1 g/mL sample of NNG3 in a 1 dm sample cell was found to have a specific rotation of −350° at 20 °C using 589 nm light.

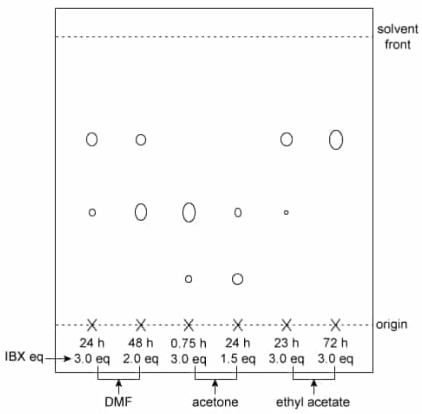


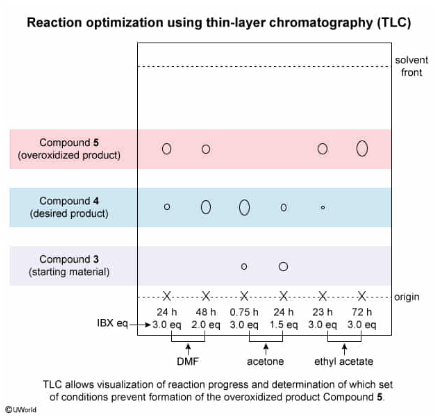
Figure 2 TLC results for oxidation reaction optimization with the reaction time, equivalents of the oxidizing agent IBX, and reaction solvent marked in each lane

Why was the optimization for the oxidation of Compound **3** monitored by thin-layer chromatography? The resulting TLC plate given in the passage is used to determine:

- A. which reaction conditions provide the greatest amount of Compound **4** and the least amount of Compound **5**. [59%]
- B. the percent yield of the reaction. [6%]
- C. whether Compound **4** or Compound **5** has a greater R_f value. [20%]
- D. which solvent and reaction time provide the greatest amount of Compound **5** and the least amount of Compound **4**. [14%]

Correct
59% answered correctly

Explanation:



Reaction optimization is necessary to determine under what conditions the desired product readily forms while preventing or minimizing side-product formation. Thin-layer chromatography (TLC) is a separation technique commonly used to visually **monitor reaction progress** when the starting material and products have different relative polarities. As such, TLC can be used to determine whether starting material is still present and whether desired products and/or side-products have formed.

The rate at which each compound travels up the TLC plate depends on the strength of the competing intermolecular interactions between the compound and the stationary and mobile phases. The passage states that the oxidation of Compound **3** yielded a mixture of Compound **4** (the desired product) and Compound **5** (the overoxidized side-product). TLC was then performed using a nonpolar mobile phase and a polar silica stationary phase. Compounds **3–5** have different strengths and numbers of interactions with the stationary phase: Compound **5** has the fewest and weakest interactions and migrates the fastest and farthest up the TLC plate whereas Compound **3** migrates the slowest.

The given TLC plate shows that different solvents and reaction times yield different relative amounts of Compounds **3–5**. This information can be used to determine the **optimal conditions**—those that yield the **greatest amount** of **Compound 4** (the product) and the **least amount** of **Compound 5** (the side-product).

(Choice B) Although TLC can show which conditions produce more or less of a compound, it is *not* a quantitative technique and does not allow determination of *exact* amounts of a compound present in a mixture. Therefore, the percent yield of the reaction cannot be calculated from the TLC plate.

(Choice C) An R_f value is the ratio of the distance a compound migrates up the TLC plate to the distance traveled by the mobile phase. Relative R_f values of two compounds can be compared without running TLC because the relative strengths and numbers of interactions two compounds have with silica in the stationary phase are known *previously*. A compound that has a stronger interaction with the stationary phase will have a lower R_f than a compound that has a weaker interaction with the stationary phase.

(Choice D) Optimal reaction conditions should provide the *greatest* amount of the desired product (Compound **4**) and the *least* amount of the side-product (Compound **5**).

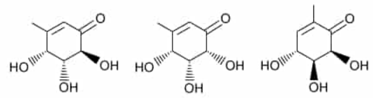
Educational objective:
Reaction optimization is necessary to determine under what conditions the product readily forms while preventing or

minimizing side-product formation. Thin-layer chromatography is a separation technique commonly used to monitor reactions because it allows visualization of the reaction's progress.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403076
System:
Introduction to Organic Chemistry

Carbasugars, also known as pseudosugars, are carbohydrate analogues in which the ring oxygen has been replaced by a carbon atom. Gabosines and their non-natural derivatives (NNG1, NNG2, and NNG3) are carbasugars with potential biological activity, including antibiotic and anticancer properties, and therefore are desirable synthetic targets.



NNG1 NNG2 NNG3

The preparation of these compounds involves hydroxyhalogenation and acetonide protection of the diol in Compound **1** to form an iodohydrin (Compound **2**). Compound **2** then undergoes an intramolecular S_N2 reaction to form an epoxide (not shown), which subsequently undergoes another S_N2 reaction with hydroxide. This is followed by the addition of the benzyl (Bz) protecting group and removal of the acetonide protecting group to form Compound **3**. Compound **3** is then oxidized using the oxidizing agent IBX to give a mixture of the desired product (Compound **4**) and an overoxidized product (Compound **5**).

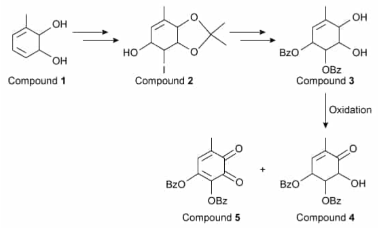


Figure 1 Selected steps in the synthesis of gabosine derivatives. (Note: Bz is an alcohol protecting group, and stereochemistry is not shown.)

Researchers studied different reaction conditions for the oxidation of Compound **3** by monitoring the reaction with thin-layer chromatography (TLC) on a silica plate with a nonpolar solvent. The results in Figure 2 show spots corresponding to Compound **4** and/or Compound **5**, depending on the reaction conditions. If the reaction did not go to completion, a spot corresponding to the starting material Compound **3** was also observed. After optimal oxidation conditions were determined, Compound **4** was converted to NNG3 by removal of the Bz protecting group. After its synthesis, a 1 g/mL sample of NNG3 in a 1 dm sample cell was found to have a specific rotation of −350° at 20 °C using 589 nm light.

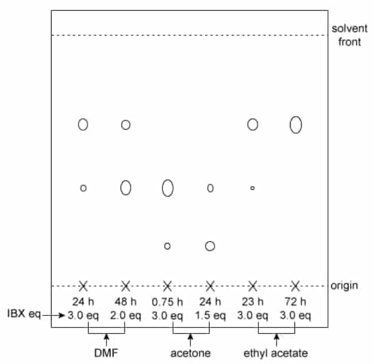


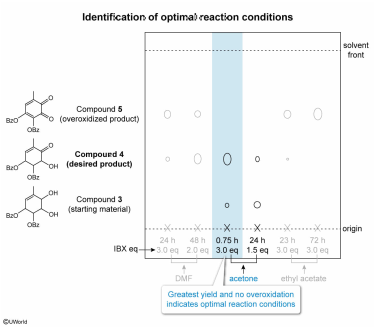
Figure 2 TLC results for oxidation reaction optimization with the reaction time, equivalents of the oxidizing agent IBX, and reaction solvent marked in each lane

Based on the TLC plate shown in Figure 2, the best yield of Compound **4** results from which reaction conditions?

- A. 48 h in DMF with 2.0 eq of IBX [9%]
- ✔ B. 0.75 h in acetone with 3.0 eq IBX [53%]
- C. 24 h in acetone with 1.5 eq IBX [9%]
- D. 72 h in ethyl acetate with 3.0 eq IBX [27%]

Correct
53% answered correctly

Explanation:



The passage states that the reaction conditions for the oxidation of Compound **3** are optimized and monitored by thin-layer chromatography (TLC) to determine the best conditions for formation of Compound **4** (the product). The optimal conditions do not further oxidize Compound **4** into the side-product, Compound **5**.

The passage states that TLC was performed using normal-phase conditions; therefore, the compound with the weakest interaction with the stationary phase migrates the fastest up the TLC plate, and the compound with the strongest interaction with the stationary phase migrates the slowest. Compounds **3–5** have the following interaction strengths with the –OH groups on the silica stationary phase:

- Compound **5**
  - Two carbonyls (C=O) act as hydrogen bond acceptors
  - Weakest interactions with the stationary phase and fastest migration (top spot)

- Compound **4**
  - One hydroxyl group (–OH) acts as both a hydrogen bond donor and acceptor
  - One C=O acts as a hydrogen bond acceptor
  - Intermediate interaction and migration (middle spot)

- Compound **3**
  - Two hydroxyl groups (–OH) act as both hydrogen bond donors and acceptors
  - Strongest interaction and slowest migration (bottom spot)

The TLC plate shows that the reactions in acetone are the only ones in which Compound **4** is not overoxidized. Because the **reactions** used the **same initial concentrations of Compound 3**, the **relative amounts** of Compound **4** formed and Compound **3** remaining can be *qualitatively* assessed by **comparing the size** of the **corresponding spots**. For the reaction that was run for 0.75 h, the Compound **4** spot is larger and the Compound **3** spot is smaller compared to the corresponding spots for the reaction that was run for 24 h **(Choice C)**.

Therefore, the reaction that was run for **0.75 h in acetone** with **3.0 eq IBX** gave the greatest yield for the oxidation of Compound **3**.

(Choice A) Compared to 0.75 h in acetone, these conditions yield a *smaller* spot for Compound **4** as well as a spot for Compound **5**, indicating some of the product was overoxidized.

(Choice D) These conditions yield only Compound **5**, indicating that all the Compound **4** that formed was overoxidized.

Educational objective:
The relative amounts of a product formed in different trials of the same reaction can be assessed by comparing the size of the spots on a thin-layer chromatography plate if the reactants all have the same initial concentration.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403077
System:
Introduction to Organic Chemistry